Lab07:
6 BT tính toán

$$X = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$W^Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \quad W^K = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix} \quad W^V = \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 1 & 1 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}$$

$$Q = XW^Q = \begin{bmatrix} 2 & 0 \\ 0 & 2 \\ 2 & 2 \end{bmatrix} \quad K = \begin{bmatrix} 0 & 2 \\ 2 & 0 \\ 2 & 2 \end{bmatrix}$$

$$V = \begin{bmatrix} 3 & 3 \\ 0 & 0 \\ 3 & 3 \end{bmatrix}$$

$$K^T = \begin{bmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \end{bmatrix}$$

$$QK^T = \begin{bmatrix} 0 & 4 & 4 \\ 4 & 0 & 4 \\ 4 & 4 & 8 \end{bmatrix} \quad S = \begin{bmatrix} 0 & 2.828 & 2.828 \\ 2.828 & 0 & 2.828 \\ 2.828 & 2.828 & 5.657 \end{bmatrix}$$

$$A = \text{softmax}(S) = \begin{bmatrix} 0.03 & 0.485 & 0.485 \\ 0.485 & 0.03 & 0.485 \\ 0.053 & 0.053 & 0.894 \end{bmatrix}$$

$$Z = AV = \begin{bmatrix} 1.545 & 1.545 \\ 2.910 & 2.910 \\ 2.841 & 2.841 \end{bmatrix}$$

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10.

pos = 0 : PE(0,0) = sin(0/1) = 0

PE(0,1) = cos(0/1) = 1

PE(0,2) = 0

PE(0,3) = 1

pos = 1 : PE(1,0) = sin(1/1) = sin(1)

PE(1,1) = cos(1)

PE(1,2) = sin(0,01)

PE(1,3) = cos(0,01)

pos = 2 PE(2,0) = sin(2)

PE(2,1) = cos(2)

PE(2,2) = sin(0,02)

PE(2,3) = cos(0,02)

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14.

$$W_1 x = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \qquad W_1 x + b_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$h = \text{ReLU}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

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